

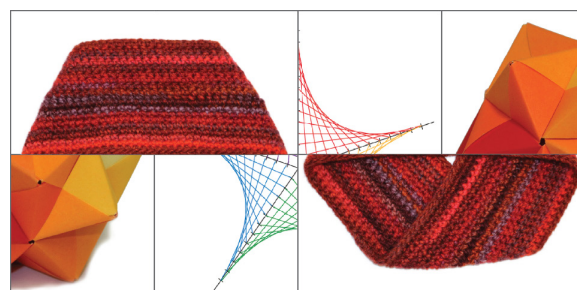
How to Fold a Tetratetraflexagon

A **tetratetraflexagon** is a folded paper polygon with four edges of equal length, like a square. But unlike a square which has two faces, a tetratetraflexagon can be flexed to reveal four faces.

To make a tetratetraflexagon, start by cutting out one of the templates provided, shown in the diagram on the left below. Pre-crease each of the grey lines (shown in the diagram on the left below) as precisely as possible. Then fold the shape in half along the middle horizontal line, and glue the wrong sides together. You should now have a rectangle which matches the two diagrams on the right below.

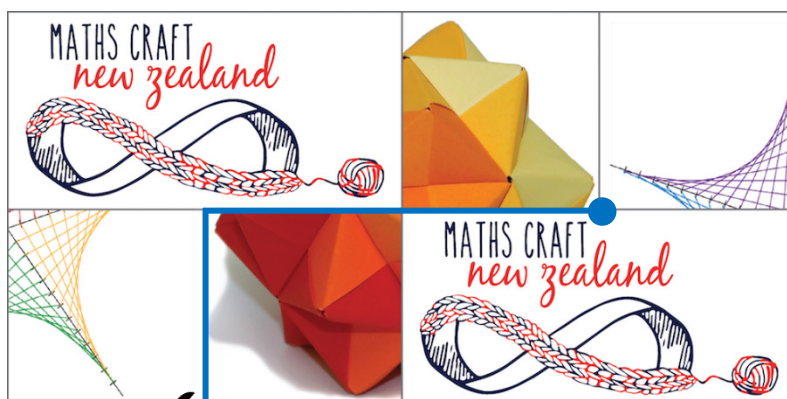


Front View



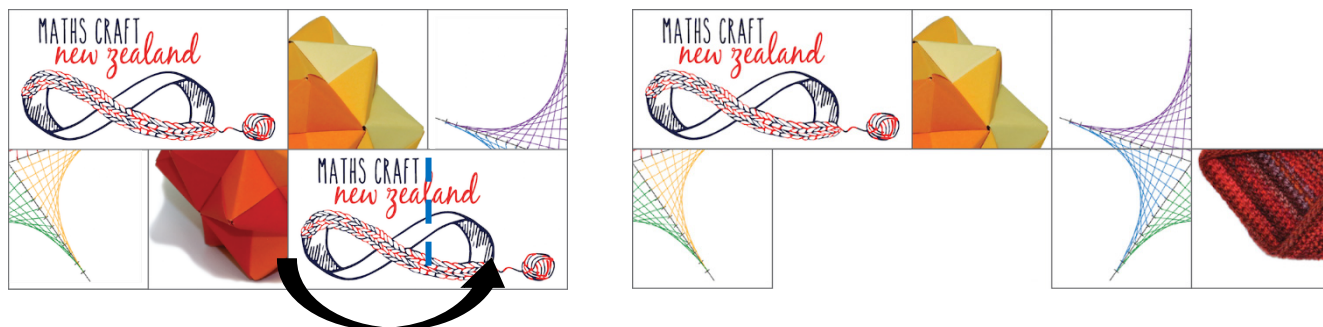
Back View

Next, with front of the rectangle facing you, cut along the solid line shown below, stopping at the dot. This creates a small flap.

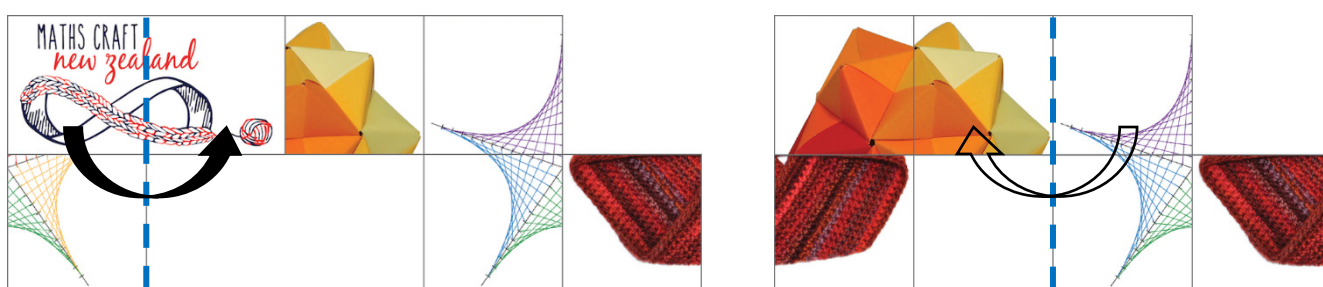


Front View

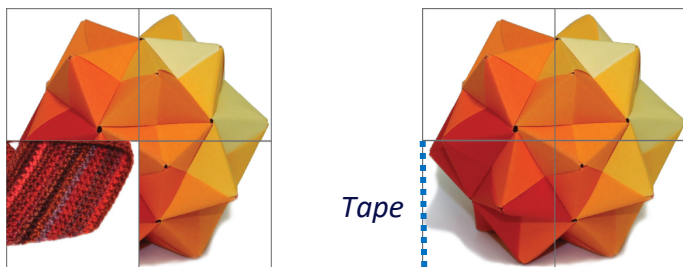
Now we'll fold the rectangle into a square. Starting with flap you have just cut, fold inwards along the dotted line shown in the diagram below left. You will obtain the shape in the diagram below right.



Next, fold the left-hand side of the shape inwards along the dotted line as shown below on the left, to obtain the shape shown below on the right. Then fold the right-hand side of the shape back along the dotted line as shown in the diagram below on the right.



You should now have a folded square identical to the one in the diagram on the left below. The final step is to tuck the bottom left-hand square behind to obtain the folded square in the diagram on the right below. One face should show an origami icosahedron and the other, a crocheted Möbius strip.



To secure your flexagon, tape the left-hand side of the bottom left square as indicated by the dotted line in the image on the left. Be sure not to tape any higher.

Your flexagon is now ready to flex!

How to Flex a Tetratetraflexagon

To flex a tetratetraflexagon, fold the flexagon in half along the vertical dotted line shown in the left-most diagram below. Carefully open up the flaps either side of the line to reveal another face. Repeat this for the next face. Not all faces will open in this way, so if the flexagon does not open easily, turn it over and try the face on the back instead. See if you can find all four faces, shown below.

